University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 2, 2017

Answer all questions.

Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Question 1 [15 Marks]

a. Briefly explain the techniques below. Be sure the difference between the two problem-solving strategies in each pair is clear. [6]

   1. *divide-and-conquer* and *decrease-and-conquer*
   2. *space-and-time tradeoff* and *problem reduction*
   3. *dynamic programming ("DP")* and *DP with memory functions*

b. State whether each of the following is true or false: [5]

   1. $\Theta (n + \log n) = \Theta (n)$
   2. $O (n \log n) = O (n)$
   3. if $p \in O (n \log n)$ then $p \in O (n^2)$
   4. if $p \in \Theta (n \log n)$ then $p \in \Theta (n^2)$
   5. if $p \in \Omega (n \log n)$ then $p \in \Omega (n)$

c. Exponentiation function a^n calculates a to the power n. Show how the problem of finding a^n is tackled using each of the following techniques (1 line each): [2]

   1. brute force
   2. decrease by a constant
   3. decrease by a constant factor
   4. divide and conquer

d. Use the Master Theorem to find the complexity of *waat* below, **<u>showing your working</u>**.

   waat (x) { if x < 3 return 'a' else return waat(x/3) ++ waat(x/3) } // ++ is concatenate

   Master Theorem: If $T(n) = aT(n/b) + f(n)$ then

   $T(n) \in \Theta(n^d)$ if $a < b^d$

   $T(n) \in \Theta(n^d \log n)$ if $a = b^d$

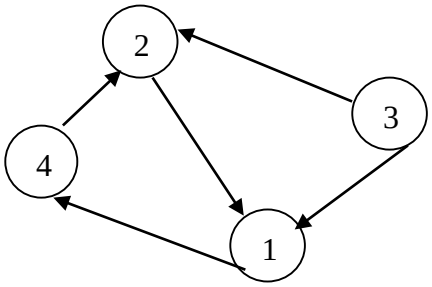
   $T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$ [2]

Question 2 [17 Marks]

a. Consider searching for the first occurrence of the octal pattern 061606 in octal text 02040606061606

1. Show the shift table that would be used if Horspool's algorithm is applied. [3]
2. Show the good-suffix table that Boyer-Moore would use for this search. [3]
3. The search starts by aligning the pattern with the first character of the text. Where will it move the pattern to next (i.e. for the 2nd alignment) using: [2]
   i. Horspool's algorithm
   ii. the Boyer-Moore algorithm
4. Can the Boyer-Moore algorithm ever have a good-suffix shift table with elements S[j] and S[k] such that j < k but S[j] > S[k]? If no, explain why not. If yes, give an example of such a search pattern. [1]
5. Which algorithm is more efficient: Horspool or Boyer-Moore? [1]

b. Consider the graph below.



1. Show the first 2 arrays (only!) that would be generated by Warshall's transitive closure algorithm (i.e. after initialising, and after the 1st iteration of the algorithm). [4]
2. What is the time complexity of Warshall's algorithm? [1]
3. What is the space complexity of Warshall's algorithm? [1]
4. Can its space complexity be improved? If yes state how, if no state why not. [1]

Question 3 [8 Marks]

a. You are in a race to find the treasure on a map. Each time you guess a point by giving its X and Y co-ordinates, you are told "YES!" or given the direction (from your guess) in which the treasure is. This direction can be North, South, East, West, North-East, North-West, South-East or South-West. The X values of the map range from -N in the West to +N in the East; the Y values of the map range from -N in the South to +N in the North.

1. Explain briefly how a brute force approach would go about solving this problem. [2]
2. Explain briefly another algorithm that can solve this problem more efficiently than your brute force one. [2]
3. What is the complexity of your brute force algorithm? [1]
4. What is the complexity of your other (more efficient) algorithm? [1]

b. The *Dutch Flag Problem* is to arrange an array of characters B, O, W (blue, orange, white) so that all the O's come first, then all the W's next, and finally all the B's last. Briefly describe a linear time algorithm for this problem. [2]